Solution 16 December, 2022

Course title: Network Optimization **Course no:** 42115

Aids allowed: All

Exam duration: 4 hours

Weighting: The points are indicated at each question

Language: You can write in Danish or English

Figures: Figures can be found on the last pages of this assignment. You can use them in your answer.

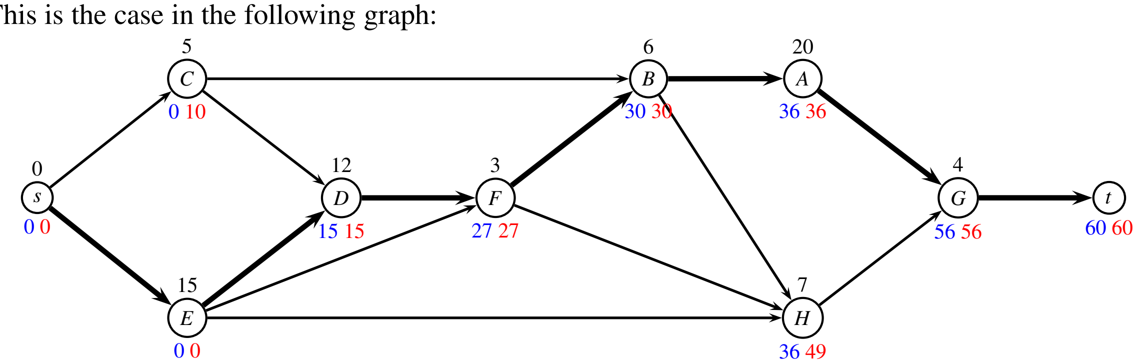
The Pisinger cake

The Pisinger cake was created by Oscar Pischinger, a famous baker from Vienna, in the 19th century.

The duration of each task is given in the following table together with predecessors that need to be finished before the task can start

task	time (min)	predecessors
A	20	B
B	6	C,F
C	5	
D	12	C,E
E	15	
F	3	E,D
G	4	A,H
H	7	E,F,B

Answer 1 A topological ordering of the activities must ensure that all edges are pointing forward. This is the case in the following graph:



■

Answer 2 Let U_i be the earliest starting time of task i (marked with blue), and let V_i be the latest starting time (marked with red).

- *Forward procedure:* Starting with $U_s = 0$, we use recursion $U_j = \max_{(i,j)\in E}(U_i + d_i)$.

- *Makespan:* $T = U_t = 60$.
- *Backward procedure:* Starting with $V_t = T$, we use recursion $V_i = \min_{(i,j)\in E}(V_j - d_i)$.

■

Answer 3 The critical path is found where $U_i = V_j$ (fat edges on above figure). ■

The following recipe is taken from Rosa Cooking
`https://www.rosacooking.com/en/pisinger-cake/`

Ingredients

- *Wafers:*
  1 pack of round wafers (you can use 8–10 home-made ice-wafers — they should be thin and flat)
- *Cream:*
  4 whole eggs
  2 egg yolks
  300 g sugar
  200 g cooking chocolate
  300 g butter
  200 g almonds
  juice of 1 lemon
- *Chocolate glazing:*
  100 g cooking chocolate
  3 tablespoons of gloss oil
  a little milk

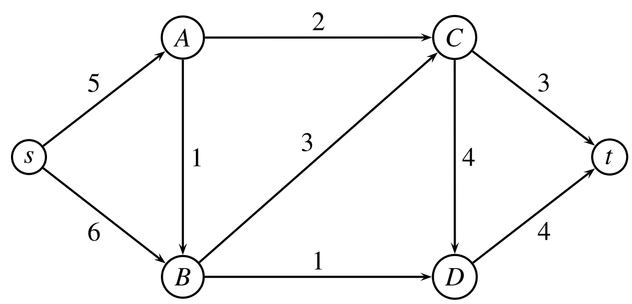
Preparation

- Using a round bowl, cut the wafers to get 10 round sheets.
- Cream: Beat eggs, chocolate, egg yolks, sugar in steam, when the mixture starts to thicken, cool it a bit, then mix in the frothy butter, lemon juice and ground almonds.
- Peel and fry almonds. Chop almonds and mix them into the cream.
- Coat the wafer-sheets with the obtained cream.
- Next leave them to cool under load.
- Pour the chocolate icing over the cooled cake and keep it cool.

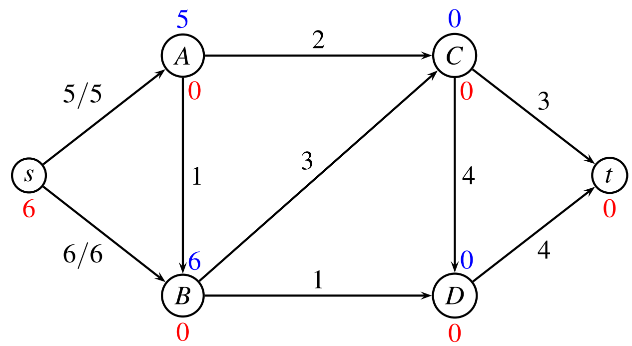
Bon appetite!

Maximum flow problem

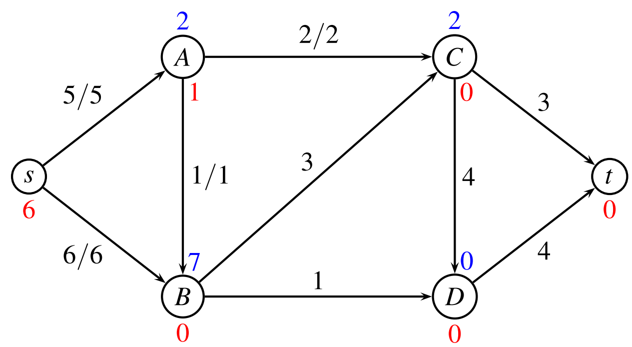
In the following network the edge weights denote the capacity. The problem is to find the maximum flow from s to t.



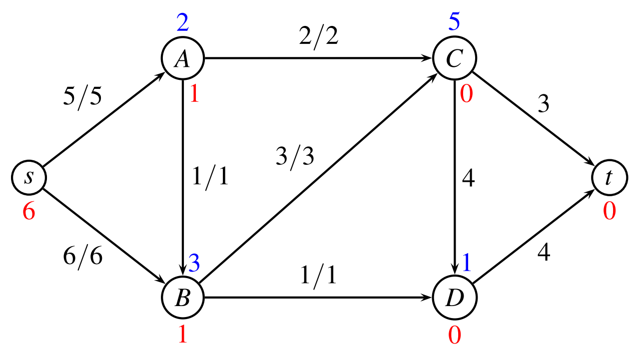
Answer 4 We use *preflow-push*. (label d_i is red, surplus $f_x(i)$ is blue)
Initialize a valid preflow by setting labels $d_s = n = 6$ and remaining $d_i = 0$. Flow on edges leaving s is set to capacity of edges.



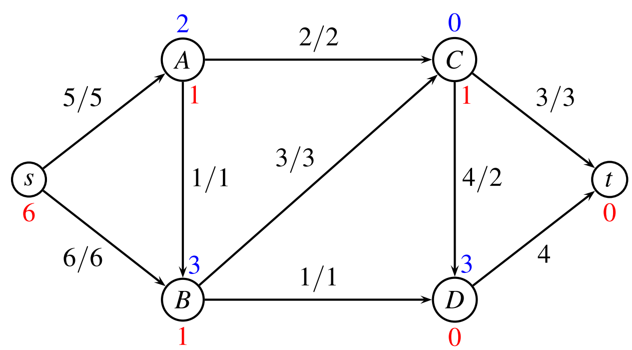
Relabel A to 1, push



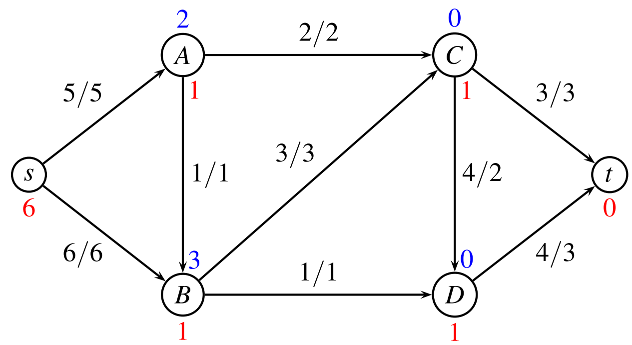
Relabel B to 1, push



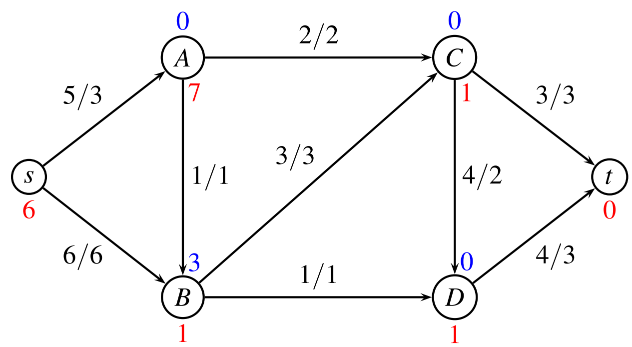
Relabel C to 1, push



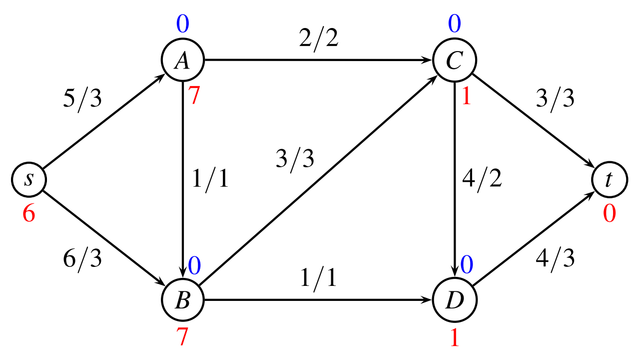
Relabel D to 1, push



Relabel A to 7 (since only outgoing edge in residual graph is As), push



Relabel B to 7 (outgoing edges in residual graph are BA and Bs), push



Since no nodes have a surplus, we have a valid flow, and hence the algorithm terminates. The maximum flow is $z = 6$.

Note: Many alterative iterations exist depending on in which order you choose the nodes. They all find the same optimal solution. **Fewest iterations:** Relabel A to $d_A = 1$, push 2 to C, push 1 to B. Relabel A to $d_A = 7$, push 2 to s. Relabel B to $d_B = 1$, push 3 to C, push 1 to B. Relabel B to $d_B = 7$, push 3 to s. Relabel C to $d_C = 1$, push 3 to t, push 2 to D. Relabel D to $d_D = 1$, push 3 to t. ■

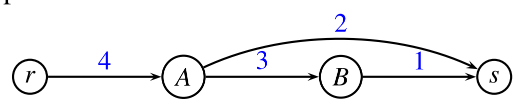
Answer 5 We use the *preflow-push* method: A minimum cut is found by choosing an unused label, e.g. $k = 2$. We then have $R = \{s, A, B\}$ and $\overline{R} = \{C, D, t\}$. The capacity of the cut is $2 + 3 + 1 = 6$. ■

Max flow min cut

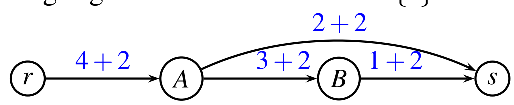
We are considering a maximum flow problem defined on a directed graph $G = (V, E)$ and edge capacities $u_{ij} \geq 0$. The solution vector is denoted (x_{ij}), and a cut is denoted $(R, \overline{R})$.

Answer 6 We have the two statements

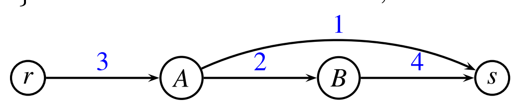
A) **NO**. The following example shows a case where the minimum cut is $R = \{rAB\}$.



Adding a constant 2 to all edges gives a minimum cut $R = \{r\}$.



B) **NO**. The following example shows a case where there are two different cuts of capacity 3: Either $R = \{r\}$ or $R = \{rA\}$. The maximum flow is also 3, so both cuts are minimum cuts.



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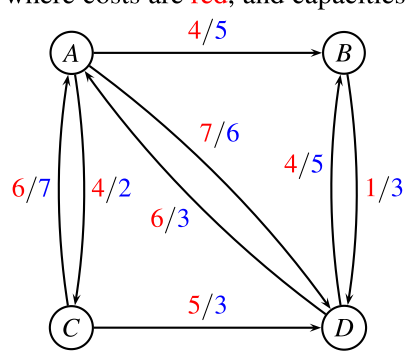
Repositioning of empty containers

Consider the following network, where costs are red, and capacities are blue.

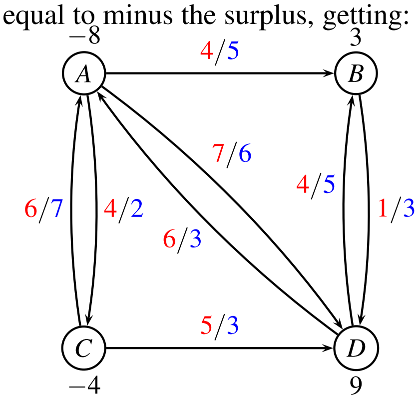


Due to the imbalance in trade, empty containers tend to accumulate at various ports. For the above network, one could have the following surplus of containers:

node i	A	B	C	D
surplus σ_i	8	-3	4	-9

We would like to reposition the containers in the cheapest possible way. It is assumed that all empty containers are equal.

Answer 7 We set the demands equal to minus the surplus, getting:



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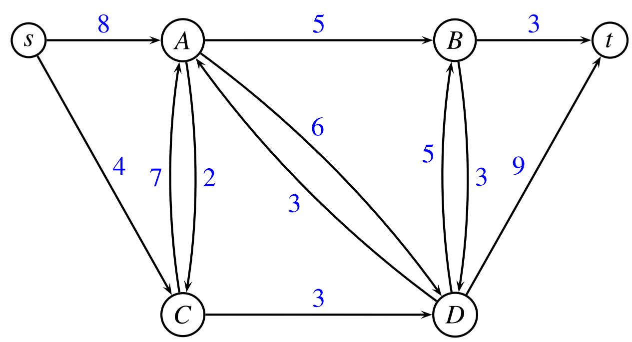
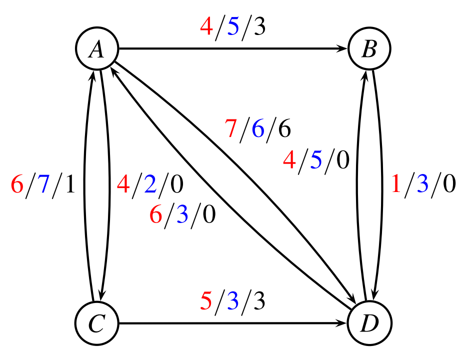
Answer 8 We add a source s and a sink t. All nodes with negative demand are connected to s with an edge having capacity equal to minus the demand. All nodes with positive demand are connected to t with an edge having capacity equal to the demand.



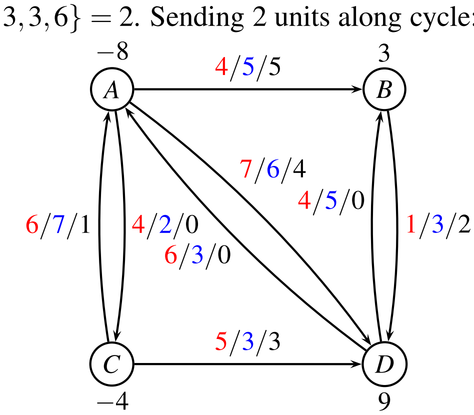
We should use augmenting paths of largest capacity, so we get:
Path $sADt$ flow 6.
Path $sCDt$ flow 3.
Path $sABt$ flow 2.
Path $sCABt$ flow 1.

We now have the initial solution



Objective value $z = 4 \cdot 3 + 6 \cdot 1 + 7 \cdot 6 + 5 \cdot 3 = 75$.
Note Alternative paths exist which may lead to a different start solution. ■

Answer 9 We use *augmenting cycle* method. An augmenting cycle is $ABDA$, cost $4+1-7=-2$. Capacity of cycle is $\min\{5-3,3,6\} = 2$. Sending 2 units along cycle:



Since no more negative cycles exist, terminate. Objective value $z = 4 \cdot 5 + 6 \cdot 1 + 7 \cdot 4 + 1 \cdot 2 + 5 \cdot 3 = 71$.

Using *Network simplex* we get: $T = \{(A,B),(C,A),(A,D)\}$ and $L = \{(A,C),(D,A),(B,D),(D,B)\}$ and $U = \{(C,D)\}$.

Using A as root we find dual variables (distances using edges in T) $y_A = 0, y_B = 4, y_C = -6, y_D = 7$. Reduced costs are found as $\bar{c}_{ij} = c_{ij} + y_i - y_j$

$$\begin{aligned}
&\bar{c}_{CD} = 5 + (-6) - 7 = -8 && (U)\\
&\bar{c}_{AC} = 4 + 0 - (-6) = 10 && (L)\\
&\bar{c}_{DA} = 6 + 7 - 0 = 13 && (L)\\
&\bar{c}_{BD} = 1 + 4 - 7 = -2 && (L)\\
&\bar{c}_{DB} = 4 + 7 - 4 = 7 && (L)
\end{aligned}$$

Edge (B,D) is in L, and has most negative reduced cost. Corresponding cycle is $\{A,,B,D,A\}$. We can send $\min\{2,5,6\} = 2$ units, getting the same flow as on top of page.

Now we have $T = \{(B,D),(C,A),(A,D)\}$ and $U = \{(A,B),(C,D)\}$ and $L = \{(A,C),(D,A),(D,B)\}$. Using A as root we find dual variables $y_A = 0, y_B = 6, y_C = -6, y_D = 7$. Reduced costs are

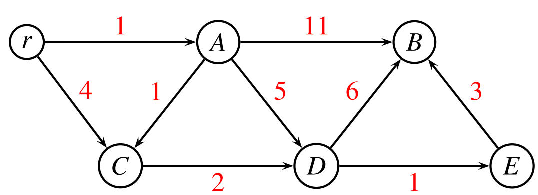
$$\begin{aligned}
&\bar{c}_{AB} = 4 + 0 - 6 = -2 && (U)\\
&\bar{c}_{CD} = 5 + (-6) - 7 = -8 && (U)\\
&\bar{c}_{AC} = 4 + 0 - (-6) = 10 && (L)\\
&\bar{c}_{DA} = 6 + 7 - 0 = 13 && (L)\\
&\bar{c}_{DB} = 4 + 7 - 6 = 5 && (L)
\end{aligned}$$

Since no edges in L have negative reduced cost, and no edges in U have positive reduced cost, we terminate. ■

Shortest Path Problem in a dynamic network

In network communication there are several protocols to ensure that packages are sent to the correct destination. One of these, the *Open Shortest Path First* (OSPF), makes use of Dijkstra's algorithm to route packages in a dynamic network, where connections are added/removed.

Consider the following directed graph where distances w_{ij} are marked with red.

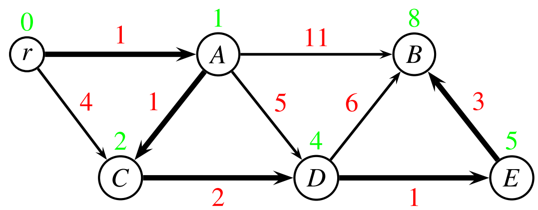


Assume that you should find the shortest path from r to all other nodes.

Answer 10 The following table indicates the nodes and their distance

considered node	r	A	B	C	D	E
init	0	∞	∞	∞	∞	∞
r	0	1	∞	4	∞	∞
A	0	1	12	2	6	∞
C	0	1	12	2	4	∞
D	0	1	10	2	4	5
E	0	1	8	2	4	5
B	0	1	8	2	4	5

Shortest path tree is:



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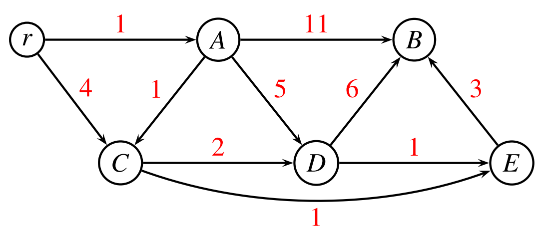
Now, assume that an edge (C,E) with distance 1 is added to the graph:

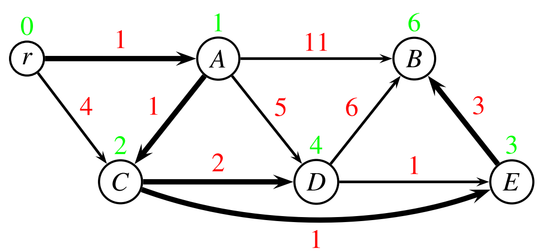


Answer 11 All distances are still valid, so we can initialize Dijkstra with the distances found in previous question. Furthermore, nodes that were considered before node C in previous question, do not need to be considered. ■

Answer 12 We initialize with the distances found in previous question, and skip nodes r and A since they were considered before node C

	r	A	B	C	D	E
init	0	1	8	2	4	5
C	0	1	8	2	4	3
E	0	1	6	2	4	3
D	0	1	6	2	4	3
B	0	1	6	2	4	3

Shortest path tree is:



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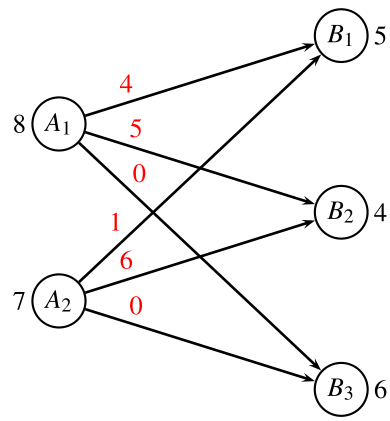
Transportation problem

Two producers A_1 and A_2 are delivering products to two customers B_1 and B_2. The producers can produce at most 8 and 7 items respectively. Customer B_1 has a demand of 5 items, while customer B_2 has a demand of 4 items. The transportation costs are given by the following table

	B_1	B_2	max production
A_1	4	5	8
A_2	1	6	7
demand	5	4	

Note, that the problem is not balanced.

Answer 13 We add a dummy node B_3 to take the excess production. Producers are connected to the dummy node with edges having cost 0.



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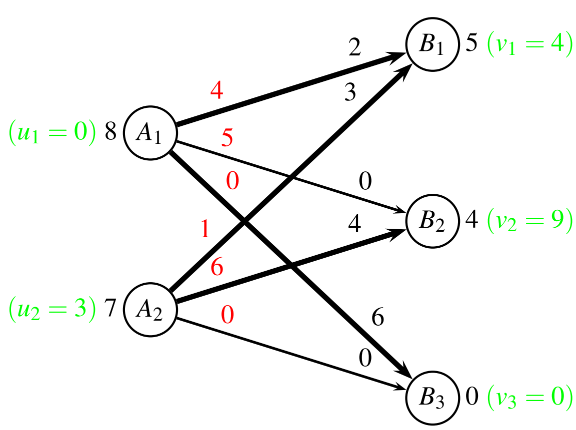
Answer 14 We first consider A_1, and send 6 units to B_3, and then 2 units to B_1. Next, we consider A_2, and send 3 units to B_1, and then 4 units to B_2. This gives the following greedy solution:



The cost of the solution is $z = 4 \cdot 2 + 0 \cdot 6 + 1 \cdot 3 + 6 \cdot 4 = 35$.

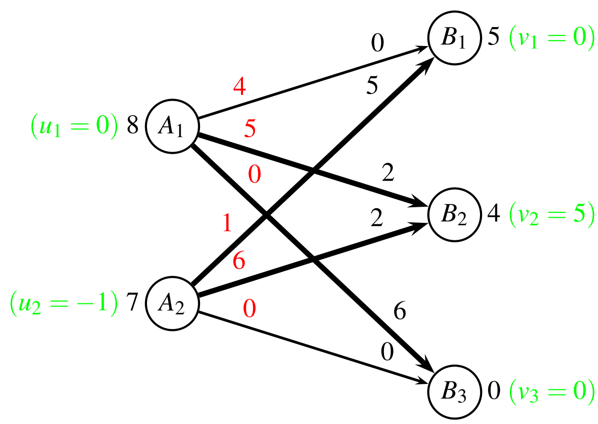
Fat edges on the graph indicate the basis. Dual values (indicated with green) are found by finding the shortest path from node A_1, using only the fat edges. ■

Answer 15 We now find the reduced costs as $\overline{c}_{ij} = c_{ij} + u_i - v_j$, getting

$$\overline{c}_{12} = 5 + 0 - 9 = -4$$
$$\overline{c}_{23} = 0 + 3 - 0 = 3$$

We choose edge (A_1, B_2) corresponding to the cycle (A_1, B_2, A_2, B_1). We can send at most $\delta = \min\{4, 2\} = 2$ units along this cycle.

After the pivot operation we get the solution

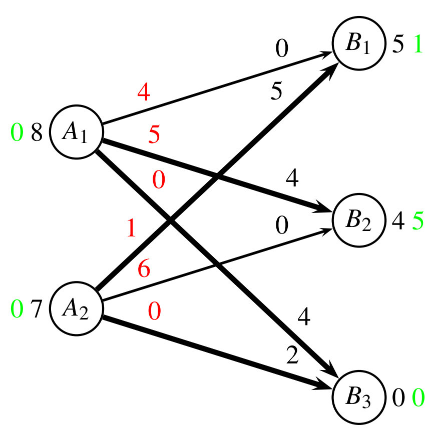


The cost of the solution is $z = 5 \cdot 2 + 0 \cdot 6 + 1 \cdot 5 + 6 \cdot 2 = 27$. Dual variables are calculated as before. The reduced costs are

$$\overline{c}_{11} = 4 + 0 - 0 = 4$$
$$\overline{c}_{23} = 0 + (-1) - 0 = -1$$

We choose edge (A_2, B_3) corresponding to the cycle (A_2, B_3, A_1, B_2). We can send at most $\delta = \min\{6, 2\} = 2$ units along this cycle.

After the pivot operation we get the solution



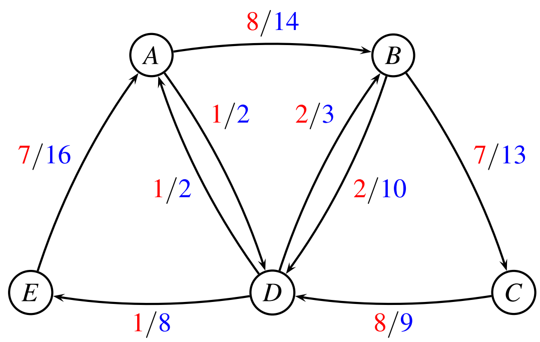
The cost of the solution is $z = 5 \cdot 4 + 0 \cdot 4 + 1 \cdot 5 + 0 \cdot 2 = 25$. Dual variables are calculated as before. The reduced costs are

$$\overline{c}_{11} = 4 + 0 - 1 = 3$$
$$\overline{c}_{22} = 6 + 0 - 5 = 1$$

No negative reduced costs, so we terminate. The optimal solution value is $z = 25$. ■

Flowing containers

MEDITLINE **(Route net of a Mediterranean liner shipping company)**



For edge (i, j) the first number is the cost c_{ij} and the second number is the capacity u_{ij} (in TEU).

We consider again the route net MEDITLINE from project assignment 2022. The network is *identical* to the one used in the project assignment, but we have *new demands* as follows:

k	s_k	t_k	d_k	w_k
1	D	C	3	2
2	A	D	2	4

Commodity $k = 1$ is shipping three 40 foot containers (size $w_k = 2$ TEU) from D to C, and commodity $k = 2$ is shipping two 40 foot high-cube containers (size $w_k = 4$ TEU) from A to D.

We are considering a *weighted multi-commodity flow problem*: For each commodity, a container of size w_k TEU has to follow the same path, but the d_k different containers can follow individual paths.

Answer 16 We have the routes r outlined in the below table. The cost of sending w_k units along a route r is calculated as $w_k \sum_{(i,j)\in r} c_{ij}$.

number r	commodity k	route r	w_k	$\sum c_{ij}$	cost
1	1	$DABC$	2	16	32
2	1	DBC	2	9	18
3	1	$DEABC$	2	23	46
4	2	$ABCD$	4	23	92
5	2	ABD	4	10	40

■

We solve the LP-relaxed problem through *column generation*. After a few iterations we have the *restricted master problem* on the form

$$
\begin{array}{lccccccccl}
\min & c_1x_1 & + & c_2x_2 & + & c_3x_3 & & & & \\
\text{s.t.} & & & 2x_2 & + & 4x_3 & \leq & 14 & & (y_{AB}) \\
 & & & & & & \leq & 2 & & (y_{AD}) \\
 & 2x_1 & + & 2x_2 & & & \leq & 13 & & (y_{BC}) \\
 & & & & & 4x_3 & \leq & 10 & & (y_{BD}) \\
 & & & & & & \leq & 9 & & (y_{CD}) \\
 & & & & & & \leq & 2 & & (y_{DA}) \\
 & 2x_1 & & & & & \leq & 3 & & (y_{DB}) \\
 & & & 2x_2 & & & \leq & 8 & & (y_{DE}) \\
 & & & 2x_2 & & & \leq & 16 & & (y_{EA}) \\
 & x_1 & + & x_2 & & & \geq & 3 & & (\overline{y}_1) \\
 & & & & & x_3 & \geq & 2 & & (\overline{y}_2) \\
 \multicolumn{6}{l}{x_1, x_2, x_3 \geq 0} & & & &
\end{array}
\tag{1}
$$

Let y_{ij} denote the dual variable corresponding to the capacity constraint for edge (i,j), and let $\overline{y}_k$ denote the dual variable corresponding to the demand constraint for commodity k.

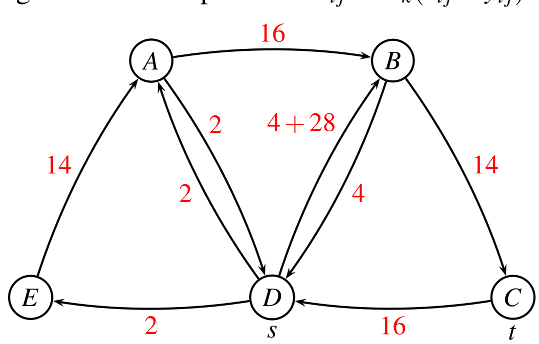
The dual variables for the above model are: $y_{DB} = -14$, while all other are $y_{ij} = 0$. Furthermore, $\overline{y}_1 = 46$, $\overline{y}_2 = 40$.

Answer 17 The reduced cost of a route r is calculated as

$$\overline{c}_r^k = \sum_{(i,j)\in r} w_k c_{ij} - \sum_{(i,j)\in r} w_k y_{ij} - \overline{y}_k = \sum_{(i,j)\in r} w_k (c_{ij} - y_{ij}) - \overline{y}_k$$

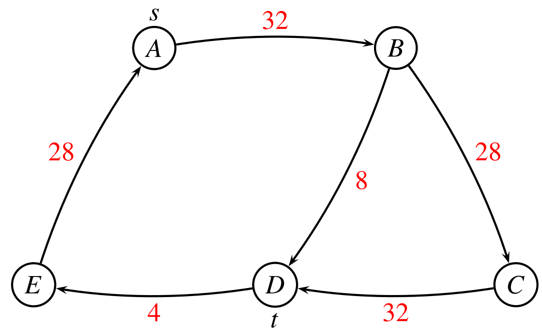
In order to solve the pricing-problem as a shortest path problem we set the cost of edge (i,j) to $\tilde{c}_{ij} = w_k(c_{ij} - y_{ij})$. For commodity k we first remove all edges with capacity $u_{ij} < w_k$. as they cannot accommodate the requested number of TEU. Next, we solve the shortest path problem using edge costs $\tilde{c}_{ij}$, with root s_k and destination t_k. let z_k be the length of the shortest path for commodity k. After solving the shortest path problem for each k, we find the reduced cost as $\overline{c}^k = z_k - \overline{y}_k$.

$k = 1$: We do not remove any edges. Costs are updated as $\tilde{c}_{ij} = w_k(c_{ij} - y_{ij})$



Shortest path is *DABC* with cost $z_1 = 32$. The reduced cost becomes $\overline{c}^1 = z_1 - \overline{y}_1 = 32 - 46 = -14$.

$k=2$: We remove edges $(A,D),(D,A),(D,B)$. Costs are updated as $\tilde{c}_{ij} = w_k(c_{ij} - y_{ij})$



Shortest path is ABD with cost $z_2 = 40$. The reduced cost becomes $\overline{c}^2 = z_2 - \overline{y}_2 = 40 - 40 = 0$.

Note: Instead of multiplying all edge weights by w_k one can also use the original edge weights, and calculate the reduced cost as $\overline{c}^k = w_k z_k - \overline{y}_k$. ■

Answer 18 The most negative reduced cost is obtained for $k=1$ with path $DABC$ (route 1) having reduced cost -14. ■

Chocolate factories

Two chocolate factories A and B are producing at most 500 chocolate-bars per month, each. Customer C demands 350 chocolate-bars, while customer D demands 400 chocolate-bars per month. Production costs are \$5 at factory A, while production costs are \$6 at factory B.

The factories can either deliver directly to the customers, or they can deliver through a warehouse W. Shipping costs per chocolate-bar are as follows:

From/to	W	C	D
A	1	5	6
B	2	4	5
W	-	3	2

The capacity of the shipping company is given by the following table:

From/to	W	C	D
A	400	500	350
B	200	600	350
W	-	200	200

The warehouse has a limited capacity, so at most 300 chocolate-bars can be shipped through the warehouse. However, at least 80 chocolate-bars need to be shipped through the warehouse.

Answer 19 We construct a network as follows: Nodes correspond to; producers A,B, customers C,D and warehouse W. The demand for producers is -500, and the demand for customers is 350 and 400.

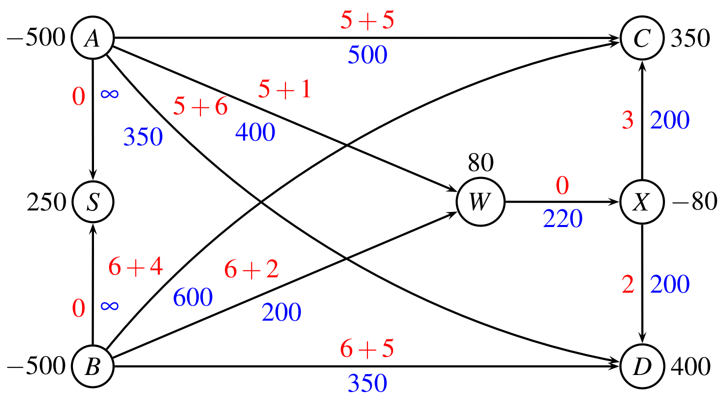
We connect each producer to the three possible nodes C, D and W. Edge costs are set to transportation cost plus production cost. Capacity corresponds to the capacity of the shipping company.

As the production of 500+500 exceeds the demand of 350+400, we add a sink node S to absorb the excess production, with demand $b_S = 500 + 500 - (350 + 400) = 250$. The producers are connected to S with edges having cost 0 and infinite capacity.

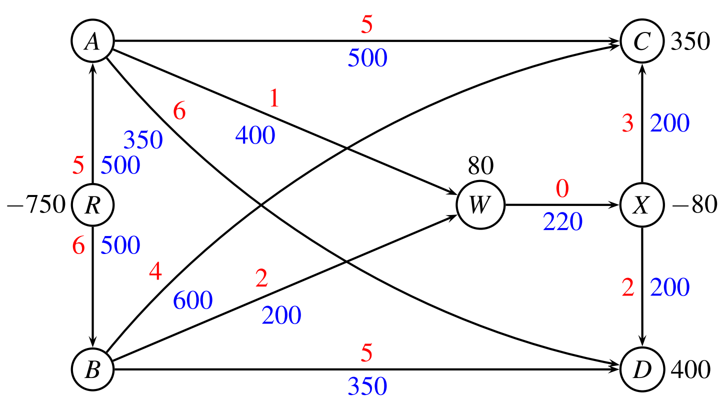
Since the capacity of the warehouse is 300, we split it into two nodes W and X, connected by an edge with cost 0 and capacity 300. There is also a lower bound ℓ on the edge, so we use the following transformation for edge (W,X)

$$u_{WX} := u_{WX} - \ell_{WX} = 300 - 80 = 220 \qquad b_W := b_W + \ell_{WX} = 80 \qquad b_X := b_X - \ell_{WX} = -80$$

Note, that since the edge (W,X) has cost 0, the objective function is not affected by the transformation, and we do not need to do any calculations after the problem has been solved.



Alternative solution Add a root node R with capacity equal to all demand, i.e. $b_R = -(350+400) = -750$. Production costs/capacities are assigned to edges (R,A) and (R,B).



Alternative solution Some students have understood the formulation, such that A and B *together* can produce 500 units. In this case the root node R above will have demand $b_R = -500$, and we need an extra node S with demand $b_S = -250$ to supply nodes C and D with the remaining demand, at no cost.
■

end of assignment